Sebastian Newberry

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Education

Wayne State University — Detroit, MI

Bachelor of Science in Computer Science | Computer Science & University Honors | GPA: 3.97 — *Aug. 2022 – Dec. 2025*

Experience

Sticker Blitz — Warren, MI

Web Developer — *Jan. 2026 – Present*

- Enhanced an existing sticker proofing AI implementation with the Vercel AI SDK, building a workflow that detects and extends colored borders on user-uploaded images, then generates precise cutlines on their artwork to turn an image into a sticker.
- Improved UX across multiple site features and supported launches of new products, including T-shirts.

Wayne State University Computer Science Capstone — *Sep. 2025 – Dec. 2025*

- Worked with three other students to develop a full-stack web application for Sticker Blitz in a capstone project for Wayne State university, delivering AI-driven customer service responses and a comprehensive customer management interface.
- Integrated the application with Vercel for deployment, AWS Bedrock for generative AI, Redis for caching, PostgreSQL as the database, Better Auth for authentication, SendGrid for email, and Pusher WebSockets for real-time communication.

Detroit Public Schools X Rocket Companies — Detroit, MI

Tutor — *Jul. – Aug. 2023, 2024, 2025*

- Led a team of 5 students in a technology tutoring program funded by Rocket Companies.
- Taught students core cybersecurity principles in web development such as authentication token security, using HTTPS vs HTTP when browsing the web, and password hashing.
- Built an interactive WebSocket-based trivia game using Socket.io, illustrating real-time communication advantages over HTTP request/response.

Wayne State Cyber Defense Club — Detroit, MI

Windows Active Directory, View Wayne State CTF 2024 and 2025 challenges — *Sep. 2022 – Dec. 2025*

- Hardened Windows Server 2012 in the Collegiate Cyber Defense Competition (CCDC) to defend against simulated cyberattacks.
- Developed and designed web exploitation, pwn, and forensics challenges for the Wayne State University CTF competition, and used Docker and CTFd to create isolated, reproducible environments to deploy them.
- Utilized tools like Wireshark for capturing network traffic, Ghidra for decompiling binaries, and Burp Suite for web reconnaissance and exploitation.

Projects

Thesivest | *React, Tanstack Start, SST, Vercel AI SDK, Neon, PostgreSQL, Better Auth* — Dec. 2025 – Present

- Built an AI-powered investment research platform with agents that search company news and analyze social sentiment to evaluate investment opportunities, inspired by the open-source multi-agent framework TradingAgents.
- Designed the application around a single editable research report as the core workflow, enabling users to iteratively refine AI-generated findings and share completed research on the platform.
- Implemented an agentic research workflow (similar to Cursor for coding) where AI performs analysis and users guide it via follow-up questions that trigger report edits and additional research.
- Deployed the project on AWS using SST behind a Cloudflare DNS proxy.

Honors Thesis - LLMs in Cybersecurity | *Python, LaTeX, OpenRouter* — Jan. 2025 – Dec. 2025

- Authored a comprehensive thesis investigating whether open-source LLMs can replace proprietary counterparts in cybersecurity by comparing 10 frontier models from both ecosystems on CTF cryptography challenges.
- Benchmarked GPT-5.1, Claude Opus 4.5, Grok 4.1, Gemini 3.0 Pro, Kimi K2, DeepSeek R1/V3, and GLM-4.6 on a SmileyCTF 2025 challenge, measuring each model's ability to derive RSA exploits from challenge code alone.
- Demonstrated that only 3 of 10 models solved the challenge: open-source Kimi K2 Thinking and proprietary Claude Opus 4.5 and Grok 4.1 Fast, suggesting open-source models can rival proprietary ones on complex security tasks.
- Surveyed both general and cybersecurity related benchmarks (MMLU-Pro, GPQA, HLE, CyberMetric, Cybench) and analyzed CIA triad tradeoffs for deploying open-source and proprietary LLMs in security workflows.

Report on RAG (Retrieval Augmented Generation) | *Python, PyTorch, LaTeX* — Feb. 2025 - Apr. 2025

- Authored a comprehensive report on Retrieval-Augmented Generation (RAG) and how external document retrieval improves LLM query accuracy.
- Implemented a practical demo of RAG using the RAGatouille Python library. View the notebook here.

Technical Skills

Languages: Python, JavaScript, TypeScript, C++, Java, R, MongoDB, SQL (PostgreSQL, MySQL), HTML/CSS
Frameworks & Libraries: Matplotlib, Node.js, React.js, Next.js, NumPy, Pandas, PyTorch
Developer Tools: Docker, Git, Cursor, Claude Code, OpenRouter
Cloud Platforms: AWS (Amplify, Lambda, S3, Bedrock, Cloudfront, DynamoDB), Cloudflare DNS, Vercel